

Yumna Waris

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EDUCATION

Otto-Friedrich-University Bamberg

Apr. 2026 – present

M.Sc. International Software Systems Science

Bamberg, DEU

- **Coursework:** German:A1, Algorithms and Complexity Theory, Advanced Data Management, Distributed Systems

University of Karachi

Jan. 2020 – Dec. 2023

B.Sc. Software Engineering

Karachi, PAK

- **Achievements:** Salutatorian / Silver Medalist: (GPA: 3.66/4.00), Nationwide Hackathon Winner
- **Coursework:** Data Structures & Algorithms, Databases, Computer Systems, Machine Learning, Deep learning

EXPERIENCE

PayEngine

Aug. 2023 – Mar. 2026

Software Development Engineer II (SDE II / Midlevel)

Santa Monica, California, United States

- Designed and shipped **end-to-end fintech features** across **React/TypeScript interfaces, Python and JavaScript backend services**, REST APIs, and PostgreSQL data models for **merchant onboarding, risk assessment, and payment processing**, serving **50,000+ merchants** and processing **100M+ transactions per month**.
- Delivered a full end-to-end **Adyen integration (Web Drop-in tokenization, digital wallets, REST backend processing)** and onboarded **8 payment processors** and acquirers (**Elavon, TSYS, Curve Pay etc.**), cutting new-processor integration **time from 6 to 1 week**.
- Engineered **fraud, transaction-monitoring, and merchant underwriting workflows** integrating **Kount, LexisNexis, Mastercard MATCH, and TransUnion CreditVision API** with business rules and **agentic process design, reducing manual fraud review by 70%, auto-approving 70% of applications**, and cutting **decision time from 3 days to 4 hours**.
- Implemented **PCI-aligned card-data security (VGS tokenization, network tokens) across 8 payment flows** and built **subscription & billing on Kill Bill**, including a custom **Java payment plugin** and **direct Payment API (PAPI) integration** for programmatic ad-hoc payments **supporting 10,000+ active subscriptions**.
- Developed **n8n automation pipelines** and **AWS QuickSight dashboards**, cutting manual onboarding and operational-review **effort by 60% ($\approx$ 17 hours/week saved)** and delivering risk, onboarding, and operational insights to **7 internal teams**.
- Tools & Architecture: Python, JavaScript, PostgreSQL, AWS, Docker, n8n, Microservices, Event-Driven Systems.

PayEngine

Feb. 2023 – Jul. 2023

Software Developer Intern

Santa Monica, California, United States

- Assisted in debugging and resolving issues within a real-world **payments platform**, gaining hands-on experience with existing systems, workflows, and production-level engineering practices.
- Developed and improved user interface components for **payment links**, enhancing usability and supporting a smoother customer payment experience.
- Collaborated with senior engineers through pair programming on **micro-deposit features, Amazon Quicksight dashboard integrations**, and a **partner subscription module** to support recurring payment functionality.
- Tools & Architecture: JavaScript, PostgreSQL, AWS, Docker, Microservices.

SKILLS

AI & GenAI: LLMs, Generative AI, AI Agents, Agentic Workflows, RAG Systems, LangChain, LangGraph, OpenAI API, Prompt Engineering, NLP, Multimodal AI, TensorFlow, Pinecone, FAISS, Tool-based reasoning, Agentic-design patterns, LLM-based workflow automation.

Data Science: Pandas, NumPy, Scikit-Learn, TensorFlow, Data Cleaning, Feature Engineering, Model Evaluation, Data Visualization.

Backend, Databases & APIs: Python, JavaScript, Java, SQL, C#, Reactjs, Nodejs, Express.js, GraphQL, Apache Kafka, Spring Boot, PostgreSQL, MongoDB, Async Processing, Git, Bitbucket

Cloud & Infrastructure: AWS (EC2, S3, Amazon QuickSight), Docker.

Software Engineering & Practices: Microservices Architecture and Design patterns, Multi-tenant Architecture, Clean Code, Scalable System Design, API-first Development Platforms, Prototyping, Technical Documentation, Stakeholder Communication.

CERTIFICATIONS

Machine Learning Specialization - Andrew Ng

Agentic AI – DeepLearning.AI

Fundamentals of Backend Engineering - Udemy

Mastering PostgreSQL: Real Examples - Udemy

Machine Learning & AI Projects

AskDB Agent – Hackathon Project

- Built an AI-powered database assistant for answering natural language questions over structured data.
- Implemented RAG-based workflows, agent orchestration, and context-aware query handling.
- Used LLMs to convert user questions into database-aware reasoning and response generation
- Tech Stack: Python, LangChain, LangGraph, OpenAI APIs, RAG

LearnScope – AI Academic Assistant

- Developed an AI-enhanced academic assistant for professors and students.
- Automated generation of course outlines, lecture materials, assessments, and academic content.
- Integrated LLM-powered workflows using LangChain and LangGraph
- Tech Stack: Python, Flask, LangChain, LangGraph, Generative AI

What Drives Exam Scores? A Clean, Reproducible ML Pipeline

- Built a clean, reproducible machine learning pipeline for tabular data analysis.
- Performed exploratory data analysis, feature importance analysis, and model evaluation.
- Tech Stack: Python, Pandas, Scikit-Learn, Data Visualization

Full Stack Projects

Concurrent Log Analysis Framework

- Designed and implemented a custom thread-pool executor and bounded blocking queue from scratch using Java concurrency primitives. Built a scalable log processing framework capable of analysing 10,000+ log files concurrently through configurable worker pools and producer-consumer architecture. Developed parallel log parsers and analytics modules to extract error statistics, keyword matches, and operational metrics from large datasets. Conducted benchmarking experiments across varying thread counts, achieving up to 5× throughput improvement over sequential execution. Implemented comprehensive JUnit test suites covering concurrency correctness, task execution guarantees, queue behaviour, and stress scenarios. Generated performance reports and benchmark visualisations to evaluate scalability, throughput, and execution efficiency.
Tools & Architecture: Java, Object-Oriented Programming, Multithreading, Concurrency, Executor Framework, Producer–Consumer Pattern, JUnit 5, Maven, Performance Benchmarking.

Nexprobiz

- Built a **full-stack corporate accounting application** for managing financial data, account books, transactions, and financial statements. Developed **complex data tables, forms, validation, filtering, and reusable frontend components** using **TypeScript, React, Material UI, React Tabulator, Formik, and Yup**, with **NestJS APIs and PostgreSQL**.